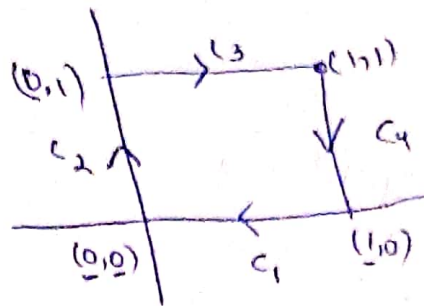


Integrate $\operatorname{Re}(z^2)$

along ~~vector~~ rectangle
with vertices $0, i, 1+i, 1$



$$C_1: z(t) = (1+t) + 0i \quad 0 \leq t \leq 1$$

$$\dot{z}(t) = -1$$

$$|z(t)|^2 = x^2 + y^2 = (1-t)^2 + 0$$

$$\Rightarrow \int_{C_1} \operatorname{Re}(z^2) dz = \int_0^1 (1-t)^2 (-1) dt$$

$$= - \int_0^1 (1^2 + t^2 - 2t) dt = - \left[t + \frac{t^3}{3} - \frac{2t^2}{2} \right]_0^1$$

$$= - \left[1 + \frac{1}{3} - 1 \right] = -\frac{1}{3}$$

$$C_2: z(t) = 0 + i2t \quad 0 \leq t \leq 1$$

$$\int_{C_2} \operatorname{Re}(z^2) dz = \int_0^1 (0^2 - t^2) i dt = -\frac{2}{3}$$

$$C_3: z(t) = t + i1 \quad 0 \leq t \leq 1$$

$$\int_{C_3} \operatorname{Re}(z^2) dz = \int_0^1 (t^2 - 1^2) 1 dt = \frac{1}{3} - 1 = -\frac{2}{3}$$

OR $z(t) = t + 0i$
t is from 1 to 0

$$\int_{C_1} \operatorname{Re}(z^2) dz = \int_1^0 (t^2) dt$$

$$= \left[\frac{t^3}{3} \right]_1^0 = -\frac{1}{3}$$

$$= - \left[t + \frac{t^3}{3} - \frac{2t^2}{2} \right]_1^0$$

$$= - \left[1 + \frac{1}{3} - 1 \right] = -\frac{1}{3}$$

$C_1:$

$$z = 1 + (1-t)i \quad 0 \leq t \leq 1$$

$$= \int_0^1 (1^2 - t^2) (i) dt$$

$$= \left(\frac{t^3}{3} - \frac{t^2}{2} \right) = -\frac{2}{3}i$$

$C_2:$

$$z(t) = 1 + it$$

t is from 1 to 0

$$= \int_1^0 (1^2 - t^2) (i) dt$$

$$= \left[t - \frac{t^3}{3} \right]_1^0$$

$$= \left(0 - 1 + \frac{1}{3} \right) i$$

$$= \frac{-3+1}{3} = -\frac{2}{3}i$$

$$\int_C \operatorname{Re}(z^2) dz = \int_{C_1} \operatorname{Re}(z^2) dz + \int_{C_2} \operatorname{Re}(z^2) dz + \int_{C_3} \operatorname{Re}(z^2) dz + \int_{C_4} \operatorname{Re}(z^2) dz$$

$$= -\frac{1}{3} + \left(-\frac{2}{3} \right) + \left(-\frac{2}{3} \right) + \left(-\frac{2}{3}i \right)$$

$$= -1 - i$$